

GRANADA AP, Spain

WMO: 084190

Lat: 37.190N

Lon: 3.789W

Elev: 1860

StdP: 13.73

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.7	28.2	17.6	14.3	38.7	21.6	17.4	38.1	20.5	49.8	17.5	50.0	1.2	210	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	33.8	100.8	68.3	98.3	67.7	95.4	66.7	72.4	93.5	70.4	91.0	68.6	89.2	9.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.2	99.9	82.8	63.2	93.1	79.5	61.5	87.5	77.6	37.3	93.9	35.3	90.8	33.8	89.1	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	15.9	13.9	21.9	105.3	3.8	2.9	19.1	107.4	16.9	109.1	14.7	110.7	11.9	112.8
			20.9	75.0	4.0	4.4	18.0	78.1	15.7	80.7	13.5	83.1	10.6	86.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	60.7	44.9	47.4	52.6	57.3	64.6	73.5	78.6	78.1	70.2	62.5	51.5	46.3	(6)
(7)		DBStd	12.97	4.64	5.08	5.03	5.53	6.33	5.61	4.28	4.34	4.99	5.44	5.90	4.91	(7)
(8)		HDD50	492	167	99	32	7	0	0	0	0	0	1	52	134	(8)
(9)		HDD65	2935	622	493	384	237	86	7	0	11	111	405	579	(9)	
(10)		CDD50	4402	9	26	113	226	453	704	888	871	606	389	97	20	(10)
(11)		CDD65	1370	0	0	0	5	74	261	423	406	167	34	0	0	(11)
(12)		CDH74	19828	0	1	55	278	1322	3783	6166	5529	2122	558	14	0	(12)
(13)		CDH80	10560	0	0	5	54	508	1997	3715	3210	926	145	0	0	(13)
(14)	Wind	WSAvg	4.3	3.2	4.2	4.7	5.2	5.2	5.4	5.2	4.7	4.0	3.4	3.3	2.9	(14)
(15)	Precipitation	PrecAvg	15.0	1.7	1.6	1.9	1.5	1.2	0.2	0.0	0.1	1.1	1.7	2.2	1.8	(15)
(16)		PrecMax	22.2	6.3	5.0	7.5	3.7	2.7	1.5	0.0	0.6	2.6	4.2	6.3	7.2	(16)
(17)		PrecMin	7.3	0.0	0.0	0.2	0.0	0.2	0.0	0.0	0.0	0.0	0.6	0.3	0.0	(17)
(18)		PrecStd	4.3	1.5	1.2	1.9	0.8	0.7	0.4	0.0	0.2	0.9	1.0	1.4	1.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.6	71.9	79.2	85.5	94.1	102.2	105.6	103.8	97.1	88.3	76.2	67.6	(19)
(20)		0.4%	MCWB	51.6	54.5	57.4	59.8	64.4	66.6	71.7	68.7	65.3	61.2	57.8	52.9	(20)
(21)		2%	DB	62.4	67.7	74.9	80.9	89.4	97.5	101.5	100.4	92.3	84.1	71.5	63.1	(21)
(22)		2%	MCWB	50.2	52.3	55.7	58.7	62.9	66.6	68.9	68.2	65.5	61.4	56.3	52.0	(22)
(23)		5%	DB	58.9	63.6	71.2	77.0	85.9	94.2	98.8	97.4	88.9	80.3	67.4	59.4	(23)
(24)		5%	MCWB	48.8	50.7	54.2	57.4	61.6	65.7	68.1	67.7	64.3	60.7	55.2	51.3	(24)
(25)		10%	DB	55.6	59.6	66.9	72.8	82.0	91.0	96.0	94.7	85.6	76.5	63.5	56.8	(25)
(26)		10%	MCWB	48.1	49.4	52.5	56.0	60.2	64.8	66.7	67.2	63.6	60.1	54.0	50.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.1	56.2	59.1	62.6	67.6	72.5	77.0	75.4	70.1	66.2	61.9	56.6	(27)
(28)		0.4%	MCDWB	61.1	67.3	75.1	79.8	87.5	92.9	99.7	96.7	88.0	79.3	69.1	61.1	(28)
(29)		2%	WB	52.6	53.9	56.7	60.1	64.7	69.5	72.8	72.4	68.2	64.2	58.7	54.5	(29)
(30)		2%	MCDWB	58.4	63.4	71.1	77.3	84.7	90.6	95.3	92.5	85.2	76.9	67.0	59.9	(30)
(31)		5%	WB	50.5	52.2	54.9	58.4	62.9	67.5	70.2	70.6	66.7	62.5	56.9	52.5	(31)
(32)		5%	MCDWB	56.5	61.0	68.2	74.5	81.8	89.2	92.5	90.3	83.3	74.9	64.3	58.2	(32)
(33)		10%	WB	48.6	50.4	53.3	56.7	61.2	65.9	68.2	68.9	65.3	61.1	54.9	50.6	(33)
(34)		10%	MCDWB	54.5	58.5	65.2	70.9	78.8	87.7	90.9	88.7	80.9	73.1	61.9	56.0	(34)
(35)	Mean Daily Temperature Range	MDBR	22.8	24.2	25.7	25.6	27.6	31.2	33.8	32.8	28.7	26.0	22.9	22.3	(35)	
(36)		5% DB	MCDBR	29.8	32.4	35.2	35.1	35.8	36.3	37.7	36.2	34.6	33.1	29.5	27.4	(36)
(37)		5% DB	MCWBR	18.4	18.7	18.5	16.3	14.8	13.4	14.5	13.0	13.1	14.0	16.1	17.3	(37)
(38)		5% WB	MCDBR	23.3	26.8	31.0	31.8	32.2	33.5	35.2	32.6	30.5	27.2	22.9	21.9	(38)
(39)		5% WB	MCWBR	15.0	16.2	16.8	15.5	14.2	14.4	16.2	14.5	12.6	12.5	13.2	14.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.298	0.326	0.366	0.364	0.381	0.400	0.391	0.398	0.377	0.361	0.324	0.304	(40)	
(41)		taud	2.517	2.452	2.333	2.367	2.349	2.309	2.330	2.315	2.395	2.431	2.505	2.522	(41)	
(42)		Ebn at Noon	285	288	284	291	286	279	281	277	277	271	272	274	(42)	
(43)		Edh at Noon	25	30	37	38	39	41	40	40	35	30	25	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	832	1085	1477	1830	2177	2478	2502	2192	1703	1239	896	743	(44)	
(45)		RadStd	88	133	151	148	167	71	55	66	103	99	84	71	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	+1.40	N/A	N/A	N/A	N/A	N/A	+153
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+137	+112

Nomenclature: See separate page